

Summary:

- Network Architect with CCIE certification, Arista Cloud Architect [ACE L7], Arista Cloud Expert [ACE L6], AWS and PaloAlto with extensive cloud integration experience
- Extensive cloud design, build experience with Azure, AWS, GCP, GCVE, AliCloud & OCI.
- Build largest network with Equinix Network Edge & Equinix Metal offering and have built multiple datacenters, Carrier hotels.
- Extended Ansible modules/collections and SaltStack State/Execution/Proxy-Minions modules to configuration management tool. [My Public Ansible Collections](#)
- Hands on with Linux, Windows, VMWARE ESXi, KVM based hypervisors, F5 LTM/GTM infrastructure
- Hands on with Python, Devops & Devsecops
- Built automation products and toolset for Cisco and Salesforce
- Presented in Cisco Live and Cisco SEVT

Certifications & Trainings:

- [Cisco Certified Internetwork Expert \[Enterprise Infrastructure\]: 43117](#)
- [Arista Cloud Expert \[ACE L7\]: AC:E7:23:AA:00:3D](#)
- [Arista Cloud Architect \[ACE L6\]: AC:E7:23:AA:00:3D](#)
- AWS Certified Advanced Networking Specialty
- Kubernetes and Docker training
- Identity Service Engine [ISE] training
- Palo Alto System Engineer certified

Technical Knowledge:

- VXLAN, EVPN across Cisco, Arista, Nokia and Juniper devices
- Firewalls: Cisco ASA, Cisco Firepower, PaloAlto, Fortigate
- Cloud: AWS, Azure, GCP, GCVE (Google Cloud VMWARE Engineer), Ali Cloud, OCI
- Hypervisors: VMWARE, OpenStack, Docker, LXC, Kubernetes
- Configuration management: Ansible, Saltstack [proficient not only using the platform but also writing modules to extend the functionality]

Patents:

- Design Patent Application #: 62807493 [SFDC: Integration of Software Application with Infrastructure]
 - ✓ Creating Infrastructure rules based on business rules
 - ✓ Intent designer

PROJECT EXPERIENCE:**Maersk Inc****Sep-2023 – Till Date****Principal Network Architect**

Maersk is the leader in transportation and is transforming itself into an end-to-end logistics provider. Technology is key for this transformation and the scope involves 250+ Maersk brands and subsidiaries across its shipping, ground, air and the logistics business. Maersk spans 1500+ land locations across globe, 650+ ships and 40+ shipping terminals that are under the administration of my team.

Roles & Achievements:**European Datacenter Migration:**

- Lead architect for European datacenter migration architecture, build and migration.
- Rearchitected the design to provide global MPLS core, modular routing stack that supports layer-2 extensions with no vendor lock-in and integration with GCP, AZURE and other cloud providers.
- Nokia based global MPLS design & deployment
- Nokia & Arista based EVPN design & deployment for DC Networking
- Design and implementation of GCP and GCVE network infrastructure and support of migration of 6000 VMs.
- Design and implementation of Azure Network design & deployment to support SAP and other business critical applications across isolated cloud infrastructure.
- Migration of services such as Zscaler & Bluecoat proxy, Firewall services and Load balancers
- Migration of DDoS service to Akamai.

APAC Integration Project:

- Planning and migration of LF Logistics business that includes 300+ logistics locations to Maersk.

- Design and build strategic datacenters to integrate multiple discrete Maersk owned business in single integrated network. This network plays an important role in Maersk's leadership vision to transform the business
- Architecture planning and POC for Enterprise Edge – a secure connectivity option for all 1000+ global offices and warehouses.
- Planning and migration of Primary European datacenters to cloud (GCP & Azure) and private datacenters such as Equinix Network Edge and Equinix Metal services.
- Solution architecture for the 30 Cloud Network Nodes (CNNs) that hosts services such as AWS, Azure, GCP and private cloud connectivity & applications such as DHCP, DNS, NTP, Wireless, Firewall and other security services.
- Planning and Migrating Network services such as Cisco AnyConnect, Cisco DMVPN, Zscaler and Akamai EA/Zero Trust

General Responsibilities:

- Develop concepts such as multi-tenants, isolated warehouses, Satellite connections for Ships, design the end-to-end solution based on business requirements and current industry trends.
- Prototype, validate the design and develop LLD, HLD and training materials for the day-2 support organization.
- Act as a bridge between architecture and automation team by setting scopes, goals and prototypes. Developed an Ansible modules that automatically validate the state of service as a short-term gap.
- Developed Ansible modules to:
 - Push configuration to Nokia and Fortigate firewalls as stop gap
 - Create Schema and Data model for Nokia Services & Firewalls and plan NetBOX implementation
 - Built Ansible based Prototype for “**Service Monitoring**” on Nokia & Arista based infrastructures to validate Interface status, BGP neighborship, Route details to confirm if the service is fully operational.

Glaxo Smith Kline/Haleon Inc.

Oct-2021 – Sep-2023

Network Architect

GSK and Pfizer created a new \$15B consumer health company named Haleon. Haleon has around 150 sites around the globe including manufacturing sites. As a migration architect my role is to plan, interact with GSK's team and build 6 carrier hotel locations in Equinix datacenters in US, Europe and Asia Pacific. Routing design and migration coordination for 150+ sites that include manufacturing sites.

Roles & Achievements:

- Lead role in design, architecture and building (6) datacenters in Equinix facility around the globe in 4 months. The data centers were built in Equinix Network Edge & Equinix Metal that are connected via Equinix Fabric.
- Azure Express Route design and implementation on all 6 sites based on new VWAN hub design
- GCP connectivity in select carrier hotels
- SDWAN connectivity with Cisco Viptela
- PaloAlto Prisma remote access solution for the remote user connectivity.
- Network build around Equinix Network Edge, Equinix Metal and Equinix Fabric
- Migration from Virtual infrastructure built around Equinix Network Edge & Equinix Fabric to physical Cisco EVPN based infrastructure
- Connectivity between datacenters via Equinix Fabric
- Network build for China at GDC facility
- SaltStack and Cisco's PYATS/Genie based, home-grown automation to improve monitoring

Forward Networks Inc.,

Feb 2020 – Oct 2021

Customer Success Architect

Forward Networks is a Silicon Valley startup company that builds a digital twin of the network by building a mathematical model from the network device state. Forward Networks can model Routers, Switches, Firewalls, Public Cloud (AWS, Azure and GCP), Hypervisors (VMWARE), WAN Optimizers (Riverbed).

Role of a Customer Success Architect is to understand the customer network, validate the network modelling, identify the critical application in the network, work with architecture, operation teams to ensure the design goal is met and improve the proactive monitoring capability, integrate to their existing tool set and support automation efforts.

Area of expertise:

- EVPN, L2VPN, L3VPN, VXLAN, MPLS
- Load Balancer
- Firewalls & VPN
- Virtualization
- Cloud Networking
- Integration

Major responsibilities & achievements:

- Act as the primary contact and liaison for customer
- Provide training, demo videos, seminars
- Build proof-of-concept automation integrating with Forward Networks

Integration Project/Forward BOT:

Forward BOT is built on SaltStack to extend Forward Networks platform and add additional capabilities. This solution has been dockerized to ease adoption.

- Built Proxy Minion for Forward Networks, Salt modules to extend the Forward Network capabilities such as scheduled health checks, self-service portal and open ServiceNow/Zendesk Tickets
- Written SaltStack engines to integrate with Slack and Microsoft Teams
- Dockerized the BOT with RestAPI based interface to administer the BOT and SaltStack

Integration Project/Ansible Modules:

Written Ansible modules to directly integrate into the configuration pipeline and health check

SalesForce Inc.,**June 2018 – Jan-2020****Network Architect, Infrastructure Engineering**

The Network Architect role involves Architecture design for first-party datacenters and define automation requirements, validate and life cycle network equipment in first-party datacenter environments.

VXLAN Migration:

SFDC.com has around 56 first-party datacenters where the production servers are grouped in multiple racks, called PODs. Architecture teams periodically upgrades the design to accommodate the industry changes. Current design is a multitier layer-2 network.

Built the next generation design with VXLAN and EVPN. Design goals

- VXLAN instead of VLAN
- EVPN
- BGP Community plan to improve route control
- Dual stack design
- Build automation template

Automated Remediation:

Hardware failure at SFDC datacenters is around 30 power supplies per month/datacenter. Operations team will replace the power supply from the spare but need open a ticket with the vendor. Goal for this project is to reduce the load by automating the process.

- Integrate with the home-grown monitoring system to identify the outage
- Integrate with Inventory system
- Open ticket with Cisco or Dell based on the hardware
- Open ticket for the operations team with the ticket and tracking details

Improve Monitoring Posture:

- TCAM provisioning based on the platform
- Configure/Validate the required modules are enabled and unwanted modules are disabled [Features]
- Configure/Validate login parameters such as banner, SSH cipher suite
- GUS ticketing for Power Supply/FAN failures along with monitoring
- POC: Power supply monitoring with SNMP

MuleSoft [a SalesForce company]**Nov 2017 – June 2018****Senior Technical Marketing Engineer [TME]**

Build a patented, Intent based automation platform, QuickSilver, from concept, design, build, and planning of the go-to-market strategy. QuickSilver, as a product has unique features of integrating the application layer or business rules to Infrastructure and supports Cisco, Juniper, PaloAlto Firewalls, VMWARE ESXi 5.5 and above, AWS (VPC and EC2) and Azure (VNET). This product was targeted at Telecom providers, ISP, and MSP markets and had a healthy \$19M pipeline and \$4M booking.

QuickSilver was built to bridge the gap between OSS/BSS to the infrastructure and provided life cycle management for the order.

This has been patented by SalesForce with me as one of three designer.

- Design Patent Application #: 62807493 [SFDC: Integration of Software Application with Infrastructure]
 - ✓ Creating Infrastructure rules based on business rules
 - ✓ Intent designer

Cisco Systems

April 2015 – Nov-2017

Solution Architect [TME]

Worked as Solution Architect/Technical Marketing Engineer with the Application Platform Group that develops applications for the Service Provider vertical. Primary responsibilities included the conceptualization & implementation of product features, building labs with new products to validate the integrations, interaction between customers and developers, training of Cisco TAC, and Advance Service organization to manage the product in the field.

Product/Bandwidth on demand: Integration between WAE and NSO/Tailf products to automate bandwidth allocation, scheduling outage for the MPLS core routers by automatically managing the traffic engineering tunnels

Product/Network Aware Workload Optimization: VM Orchestration platform that manages OpenStack, VMWARE & Docker based virtual infrastructure by integrating with VMTurbo [Turbonomics] and Tailf. Provides life cycle management [Create, Update, delete] for VM. This product is intended for service providers that are building private clouds for their customers

Technologies worked on:

- Data Center: ACI, VPC, VXLAN & Spine/Leaf architecture
- WAN: iWAN, MPLS Traffic-Engineer/TE, BGP Link-State [BGP-LS], Segment Routing
- Controllers: OpenDayLight [ODL], SDN/NFV, Network Automation [NSO/Ansible], WAE [Wan Automation Engine]
- Cloud: AWS, Azure [Networking & Compute]
- Network: Cisco [IOS, XE, XR, NX-OS & ASA], Juniper [MX, SRX] and Arista
- Virtualization: VMWARE, OpenStack, Hyper-V & Docker
- Automation & DevOPS: Ansible, Python & SALT

Automatic Data Processing [ADP]

July 2014 – April-2015

Architect

Worked as Architect in the GNS Data Networking team, responsible for Core Routing & Switching [Arista and Cisco based switches & routers], Cisco Wireless Infrastructure and B2B VPN/Remote Access VPNs [ASA & IOS/IOS-XE based]

Major responsibilities & achievements:

- Architecture of Spine/Leaf with Arista switches
- Re-engineering Cisco Wireless infrastructure with High Availability & ISE infrastructure
- Architecture for GEN4 B2B VPN infrastructure with IPSEC VPN with ASA firewall and DMVPN
- POC/Vendor evaluation for OpenStack/SDN infrastructure & finalized on Nuage Network
- Architecture ownership for Cisco based Wireless infrastructure. Re-architected controller failover, Wireless code upgrades, 1/2/3 years' migration plan for Wireless AC & Controller license

Cisco Systems

April 2012 – July 2014

Network Consulting Engineer

Worked as Network Consulting Engineer in Cisco Advanced Services [Cisco AS] and worked with multi-national customers in Financial, Insurance, & Retail sectors in around the globe

American Insurance Group [AIG]:

Societe Generale

GlobalFoundries Inc

April-2012 to July-2014

April 2012 to July-2014

April 2013 to July-2014

TD Ameritrade

October 2013 to July-2014

Short Term engagements:
Atlantic Health Services [AHS]
Novartis
ADP Inc
1-800FLOWERS:
Moody's:
Ernst & Young [E&Y]:

April 2014 to July-2014
September 2012 to Jan 2013
Multiple engagements
June-2012 – Oct-2012
June-2013 – July-2013
June-2012 – Aug-2012

MY PREVIOUS PROJECTS/EXPERIENCE:

COMPANY	PERIOD	JOB TITLE
Federal Reserve Bank	Dec '11 – Apr '12	Senior Network Engineer
Becton, Dickinson and Company	June '11 – Oct '11	Security Architect
CityTime/Office of Payroll & Administration/SAIC	Aug '08 – June '11	Senior Network Engineer
United Water Inc	Feb '03 – Aug '08	Senior System/Network Engineer